BALASANKAR P

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Summary

Data Scientist with experience building production AI/ML pipelines on Google Cloud Platform for enterprise energy clients. Work spans computer vision, time-series forecasting, geospatial analytics, and NLP. Published research at Springer 2025 on Knowledge Graph-driven AI.

Experience

Dataneurus (formerly Bulerez Consulting) **Apr 2025 – Present**
Associate Consultant – Data Science | Client: CenterPoint Energy, USA *Bangalore, India*

- Developed an AI pipeline to detect corrosion on gas meters using **Google Gemini** with few-shot in-context learning, automating inspection across **900K+ survey records** in 7 regions. Deployed on GCP using Cloud Run, Pub/Sub, and BigQuery.
- Reduced BigQuery query costs by **96%** by implementing table partitioning and clustering on large-scale image attachment tables, unblocking a stalled production data pipeline.
- Developed a risk scoring pipeline for 811 dig tickets across **10 Texas counties**, computing composite scores from geospatial features (census density, GIS pipeline attributes, proximity analysis). Implemented automated SMTP alerts for emergency tickets.
- Built an ensemble forecasting system using **Prophet**, **SARIMA**, and **LightGBM** with automated model selection per county and work-type, predicting weekly ticket volumes 4 weeks ahead to support proactive resource allocation.
- Developed a **Streamlit** headcount forecasting application using Prophet and Monte Carlo simulation to predict technician staffing across **11 service areas**. Integrated **Gemini AI** to generate plain-language explanations of staffing gaps for operations leadership.
- Built a daily GPS validation pipeline to cross-reference technician-reported arrival times against vehicle GPS data for **∼500 emergency orders/day**, achieving **89.5% automated validation rate** for regulatory compliance.
- Scoped and designed an **ML observability platform** for automated data drift detection with configurable alerting (Slack/Email/PagerDuty) and composite trust scoring for monitoring deployed model health.
- Collaborated with US-based client stakeholders on requirements gathering, sprint demos, and technical presentations across multiple concurrent projects.
- Contributed to an internal **development accelerator**, building reusable tooling for **LLM hyperparameter tuning** and **summarization evaluation suites** to standardize experimentation workflows across projects.
- Supported internal marketing efforts through technical content, solution demos, and case study contributions for client outreach.
- Migrated enterprise **SAP data to GCP BigQuery**, creating logical views to replicate table structures and business logic. Wrote and optimized SQL queries for data transformation and schema validation.

IP Ever LLP **May 2024 – Aug 2024**
Data Analyst and AI Development Intern *Chennai, India*

- Built a **RAG chatbot** integrating PostgreSQL, S3, and ChromaDB with **Mistral LLM** via Groq, using **CrewAI agents** for dynamic SQL generation from natural language queries. Served via **FastAPI** with JWT authentication.
- Developed an automated patent classification system achieving **85% accuracy** and built **FastAPI**-based report generation using **LLaMA** and **Mistral**, improving metadata processing efficiency by **40%**.

Tiger Analytics India Consulting Private Limited **Jan 2024 – Apr 2024**
Data Science Intern *Chennai, India*

- Built **Tableau dashboards** for retail KPI tracking with automated DAG pipelines running 3x daily (GCS → BigQuery). Created SQL views and published dashboards to **Tableau Cloud** for stakeholder access.

Projects & Publications

Knowledge Graph-Driven Multi-Agent Framework for Agriculture | *LLaMA 3, LangChain, Neo4j, SPARQL* **Jan 2025**

- **Presented at Springer Research Conclave 2025.** Built a KG-driven multi-agent framework integrating **Agentic RAG** for contextual information retrieval in agriculture. Preprocessed and structured data from multiple sources into knowledge graph triplets for automated reasoning.

Biomedical Knowledge Graph Construction from Literature | *BioBERT, Neo4j AuraDB, MinIE* **Nov 2024**

- Constructed a biomedical knowledge graph from the NCBI dataset using coreference resolution and MinIE for entity extraction (**98% accuracy**), enabling disease–chemical interaction queries via Cypher with a scalable ETL pipeline.

EduRecap – Intelligent Lecture Summarization App | *Whisper, LLaMA, React Native, DistilBERT* **Oct 2024**

- Developed a mobile app using **Whisper** for real-time lecture transcription (**90%+ accuracy**) and **LLaMA** for context-aware summarization, with automated off-topic content filtering reducing review time by **40%**.

Technical Skills

Languages: Python, SQL, Java, JavaScript, R, HTML/CSS
AI/ML: Computer Vision, NLP, Generative AI, Deep Learning, Agentic RAG, Knowledge Graphs, Transformers, Prophet, SARIMA, LightGBM
Cloud & Infrastructure: GCP (BigQuery, Cloud Run, Vertex AI, Cloud SQL, GCS, Pub/Sub, IAM), AWS (EC2, S3, Lambda), Docker, CI/CD (GitHub Actions)
Frameworks & Tools: FastAPI, Streamlit, LangChain, CrewAI, Neo4j, ChromaDB, Tableau, Power BI, Pandera, Git

Education

Christ University **2023 – 2025**
Master of Science in Data Science *Bangalore, India*

Amrita Vishwa Vidyapeetham **2020 – 2023**
Bachelor of Computer Applications in Data Science *Kerala, India*

Certifications

AWS Certified Cloud Practitioner | IBM Introduction to Data Engineering | IBM Data Privacy for Information Architecture